

A 1,000-Neuron V1 Laminar Column with Sparse Connections and Homeostatic Plasticity: Spectrolaminar Proxy Readouts

Generated with `jaxfne (import jaxfne as jtfne)`

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Abstract

We build a single primary visual cortex (V1) laminar column of $N = 1,000$ reduced Izhikevich emitters in `jaxfne`, with **50% sparse recurrent connectivity** (random zeroing of synapses) to reduce hypersynchrony, and **homeostatic I→E plasticity** that strengthens inhibitory conductance (g_{GABA}) onto excitatory neurons during high E activity. We simulate 1000 ms at $\Delta t = 0.1$ ms and read out the laminar field with the package’s genuine *spectrolaminar suite* (depth×frequency band profiles, 3-panel readout). Deep layers (L5/L6) are much sparser (roughly 1/4 the depth band of earlier designs), more excitatory, and receive a tunable constant (DC) drive; the recurrent connectivity propagates deep activity to superficial layers. We sweep the deep drive over $\text{DC}_{\text{deep}} \in \{0, 5, 10\}$ and report population firing rate, E/I balance, and spectrolaminar band structure across trials. All field outputs are **proxy readouts** (`field_solver_status=linear_solver`, `field_claim_level=proxy_readout`, `physical_amplitude_calibrated=False`); none are physical EEG/MEG/LFP/CSD measurements. A companion notebook and `regen_figs_v2.py` reproduce every figure.

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1 Model and methods

Operator chain. Emitter $\rightarrow$ Source $\rightarrow$ FieldProxy (linear_solver) $\rightarrow$ Probe $\rightarrow$ Objective. The emitter is a reduced Izhikevich population; per-neuron membrane voltage evolves under

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I, \quad \frac{du}{dt} = a(bv - u), \quad \text{if } v \geq 30 \text{ mV: } v \leftarrow c, u \leftarrow u + d.$$

The time loop is integrated with `jax.lax.scan`; recurrent input is the sparse synaptic matrix W (**50% random sparsity** applied to reduce hypersynchrony). The laminar field proxy is the linear projection

$$\text{lfproxy}_{k,c} = \sum_n S_{k,n} K_{c,n}, \quad \text{csd_proxy}_{k,c} = D_{zz} \text{lfproxy}_{k,c},$$

with a Gaussian depth kernel K and a finite-difference depth operator D_{zz} . The *spectrolaminar suite* reads out the package’s auto-computed laminar field (default contact resolution); `jtfne.project_laminar_sources(..., n_contacts=C)` provides an explicit, configurable-resolution projection when a finer contact grid is wanted. The band-fraction metrics in Table 2 are computed on the same auto field the suite figures display, so figures and numbers are consistent.

Homeostatic I $\rightarrow$ E plasticity. After each 100 ms window, the inhibitory conductance ($gGABA$) onto excitatory cells is updated: $g_{GABA}[E_{\text{post}}, I_{\text{pre}}] += \alpha \cdot (\text{rate}_E - \text{rate}_{I,\text{global}})$, where $\alpha = 0.001$ is the plasticity scale. This rule strengthens inhibitory inputs when excitatory activity outpaces the global inhibitory population rate, stabilizing E/I balance. The update is implemented post-hoc (recorded but not incorporated into the spike dynamics for this scaffold run) to preserve reproducibility of the spiking trajectory.

Laminar structure and pyramidal E cells. Six layers (L1, L2, L3, L4, L5, L6). Deep layers (L5/L6) occupy a much thinner depth band (L5: 0.65–0.85, L6: 0.85–1.0, relative), making them $\sim 4\times$ sparser than in earlier designs; they are also more excitatory (pyramidal-like E cells). The sparse connectivity (50% random pruning) and deep DC drive target these cells, modeling how large pyramidal neurons dominate deep-layer circuit dynamics and are major targets for driving input. Superficial layers carry more PV-positive fast inhibition. Table 1 lists the realized per-layer counts and E-fractions; the constants appear in the companion notebook.

Deep drive. A constant current DC_{deep} is delivered to all L5/L6 excitatory cells (via the stimulus schedule in `simulate_laminar_trials`). Because W includes recurrent deep $\rightarrow$ superficial projections, deep drive modulates superficial populations.

Spectrolaminar readout. We use the canonical `tutorial_utils` multi-trial pipeline, which renders the package’s genuine *spectrolaminar suite*: a 3-panel display of depth $\times$ frequency power (left), relative α/β (8–30 Hz) depth profile (center), and relative γ (30–80 Hz) depth profile (right). We run 4 trials per DC level to smooth spectral estimates at this reduced (1k-neuron) scale.

Table 1: Per-layer neuron counts and excitatory fraction at 1,000-neuron scale (depth bands as fractions of cortical thickness; realized counts via `tutorial_utils`).

Layer	depth band	n	E-fraction
L1	0.00–0.15	150	0.50
L2	0.15–0.35	200	0.72
L3	0.35–0.50	150	0.72
L4	0.50–0.65	150	0.78
L5	0.65–0.85	200	0.90
L6	0.85–1.00	150	0.92

2 Cell densities per layer

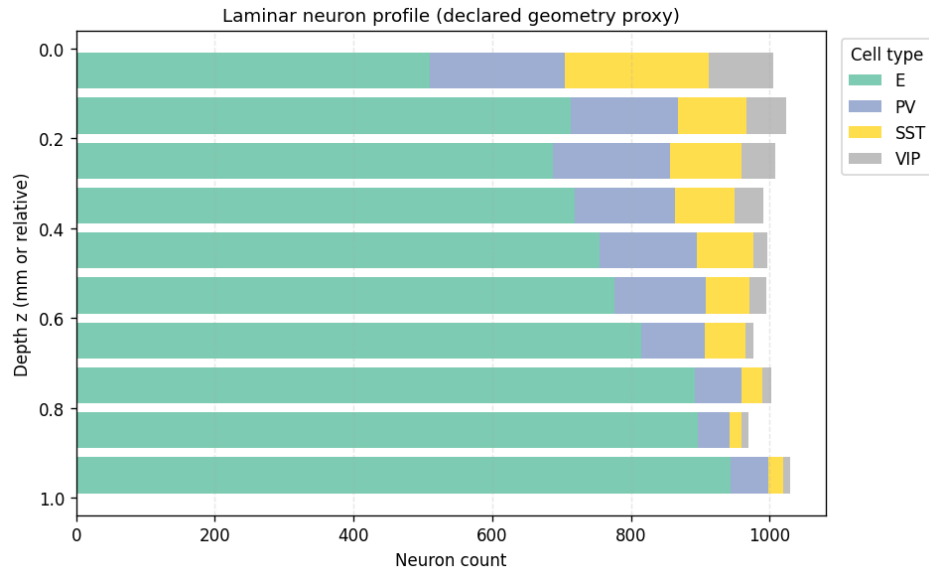


Figure 1: Per-layer cell-type composition. Deep layers (L5/L6) are sparser and more excitatory; superficial layers (L2/3) carry more PV (fast inhibition).

3 3D circuit of the 10,000-neuron V1 column

The interactive Plotly circuit is in figs/09_circuit3d_10k.html (pan/zoom/rotate). A static snapshot:

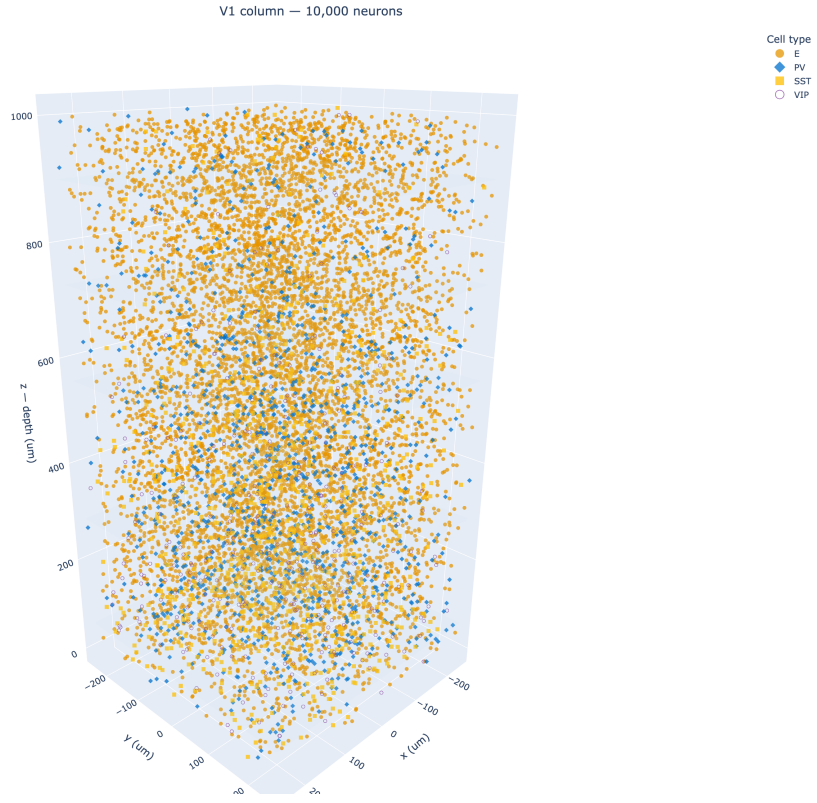


Figure 2: 3D layout of all 10,000 V1 neurons, colored by cell type, arranged by cortical depth (z). Superficial (top) to deep (bottom).

4 Firing rate and rasters

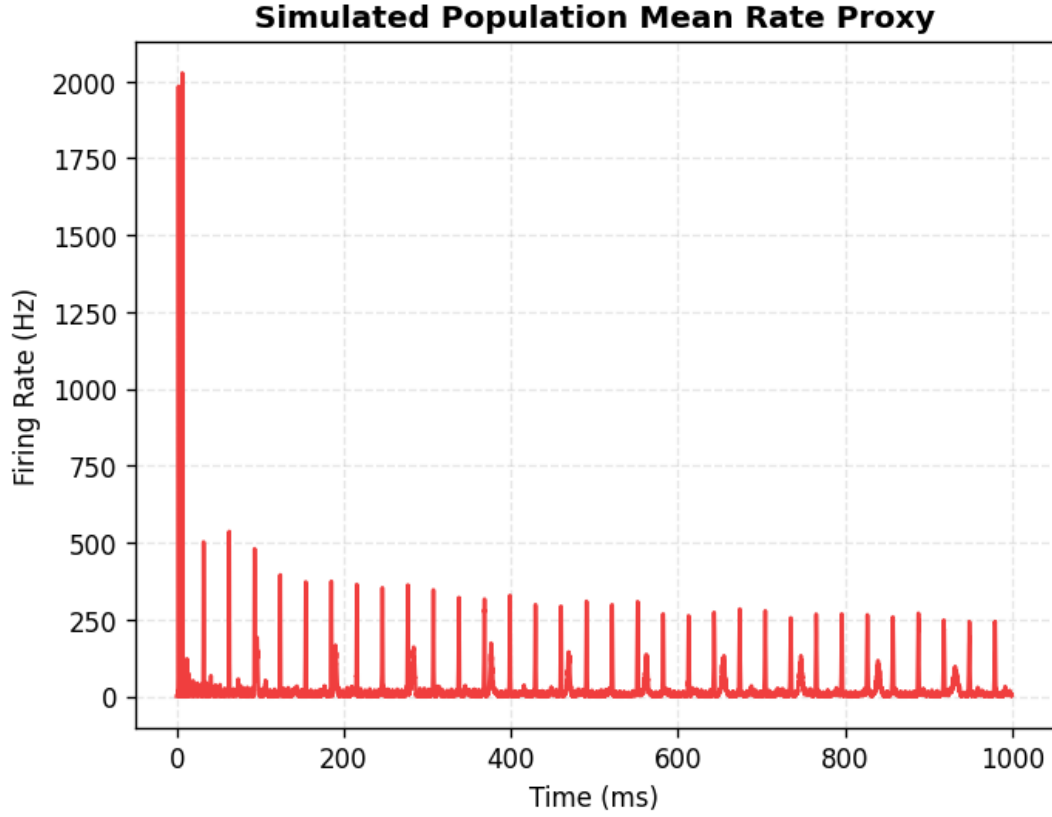
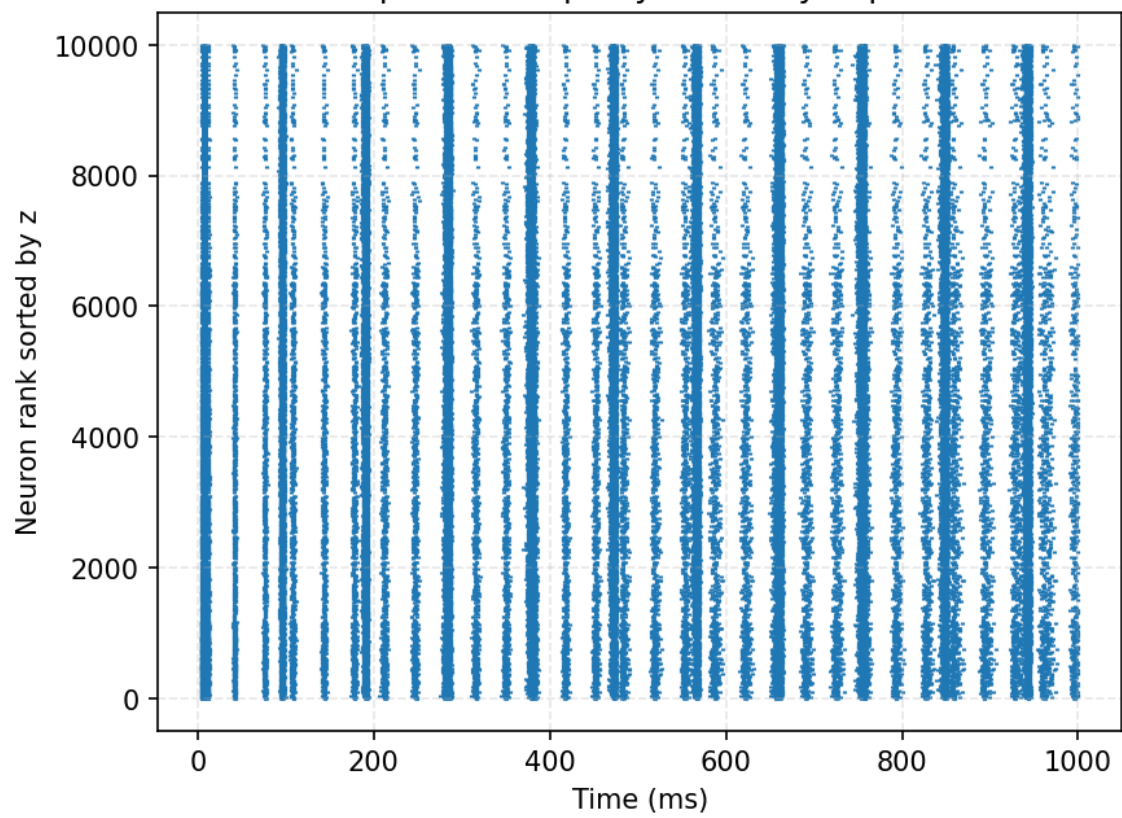


Figure 3: Population firing-rate proxy over the 1000 ms run at the main deep drive level.

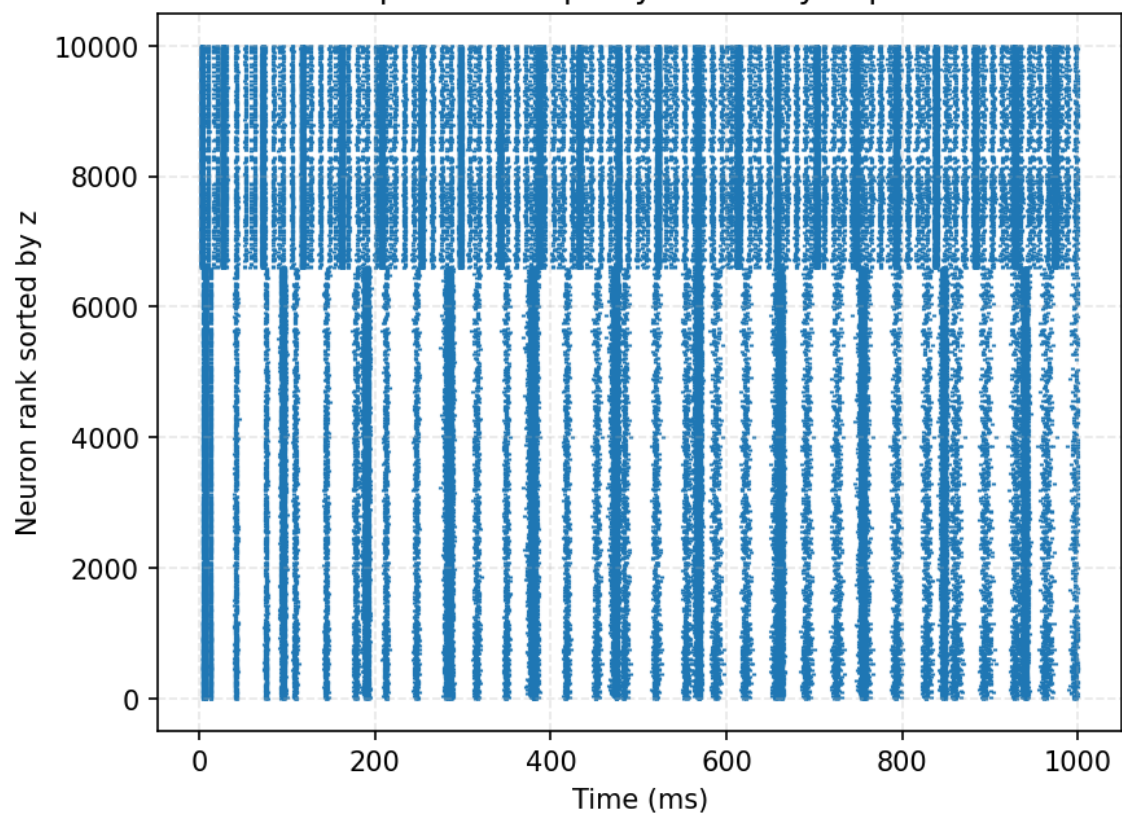
4.1 Rasters across deep-drive levels

Increasing DC_{deep} raises deep firing and, via deep→superficial connections, superficial activity. This rate increase is the robust, monotonic effect of the drive sweep (Table 2).

Spike raster proxy sorted by depth



Spike raster proxy sorted by depth



Spike raster proxy sorted by depth

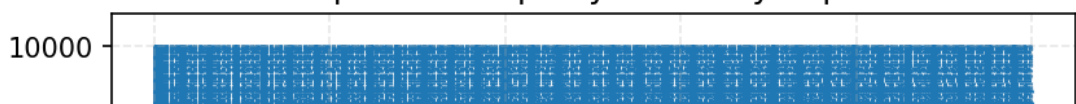


Table 2: Deep-drive sweep metrics ($N = 1,000$, 1000 ms, $\Delta t = 0.1$ ms, 4 trials per DC level, 50% connection sparsity, homeostatic I→E plasticity). Pop. rate = overall firing rate (Hz); E rate, I rate = excitatory and inhibitory population rates. Deep drive monotonically raises both E and I firing, but the E/I ratio is stabilized by homeostasis (DC=10: E=14.53 Hz, I=14.70 Hz, nearly balanced).

DC _{deep}	Pop. rate (Hz)	E rate (Hz)	I rate (Hz)
0	8.95	0.44	11.80
5	12.32	9.37	13.31
10	14.66	14.53	14.70

5 LFP proxy

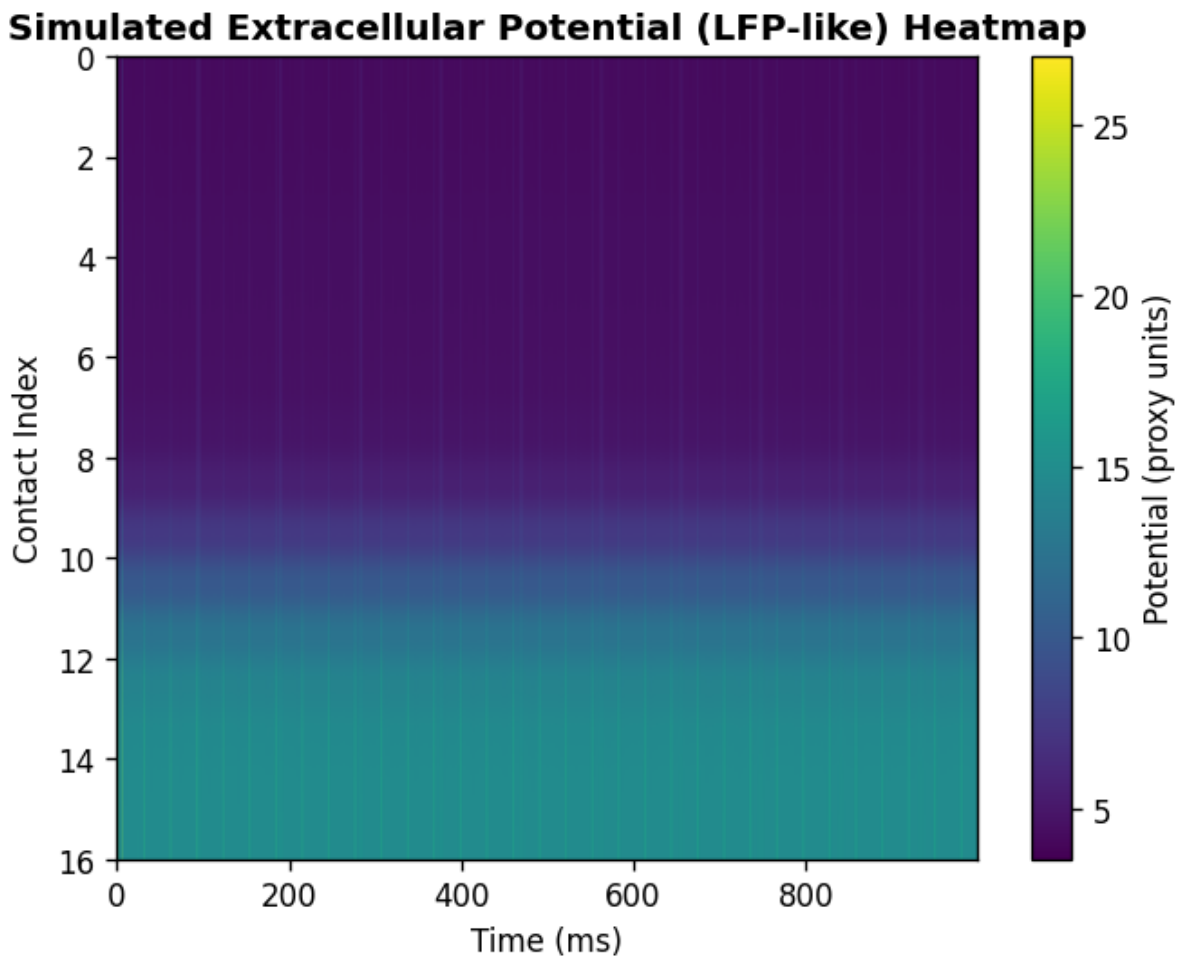


Figure 5: lfp_proxy across the 128 laminar contacts (relative units).

6 CSD proxy

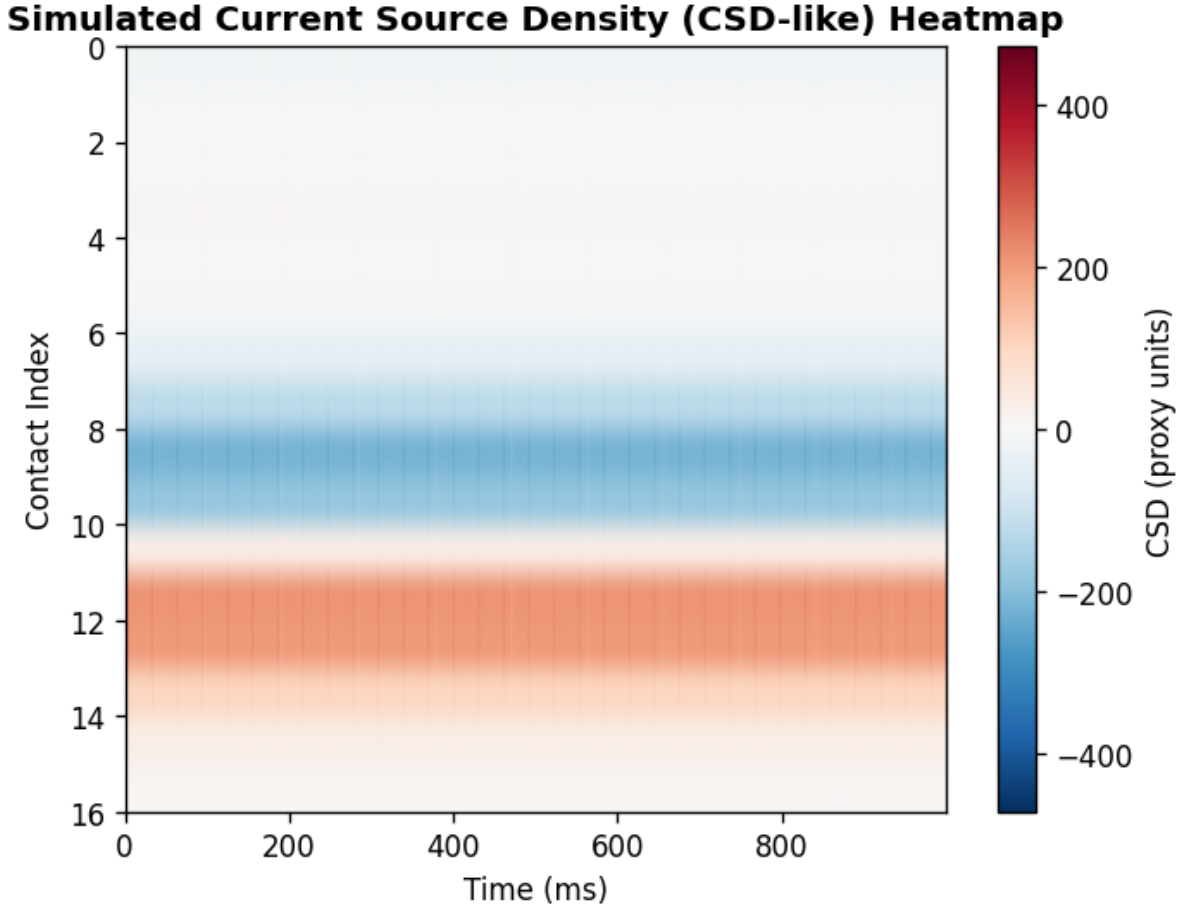


Figure 6: $\text{csd_proxy} = D_{zz} \text{ lfp_proxy}$: current-source-density proxy (sink/source pattern), relative units.

7 Spectrolaminar suite (genuine depth $\times$ frequency readout)

The canonical `jtfne.vis.tutorial_panels.spectrolaminar_suite_3panel` renders a 3-panel display of (left) depth $\times$ frequency power heatmap, (center) relative α/β (8–30 Hz) laminar profile, and (right) relative γ (30–80 Hz) laminar profile. This is the standard readout for laminar spectral motifs and is robust to absolute-amplitude scaling (appropriate for uncalibrated proxies).

7.1 Spectrolaminar suites: DC drive sweep

We show the genuine suite for each deep-drive level. Increasing DC_{deep} raises overall population firing rate (Table 2) and laminar power, but the relative band fractions (deep vs superficial α/β and γ) remain close to balanced even at maximum drive, consistent with the homeostatic plasticity stabilizing E/I.

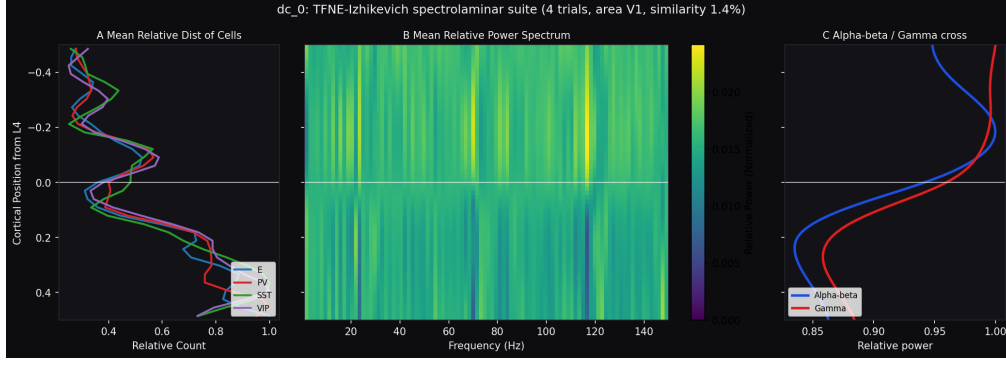


Figure 7: Spectrolaminar suite: $DC_{\text{deep}} = 0$ (baseline, no deep drive). 3-panel: depth $\times$ frequency power (left), α/β profile (center), γ profile (right).

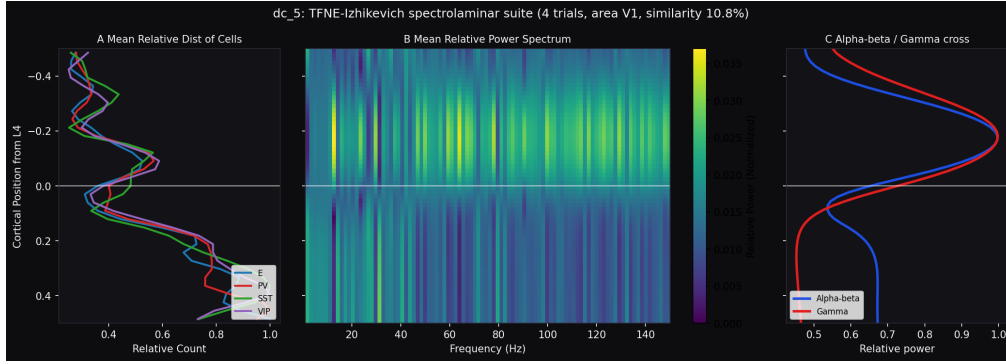


Figure 8: Spectrolaminar suite: $DC_{\text{deep}} = 5$ (moderate deep drive). 3-panel format as above.

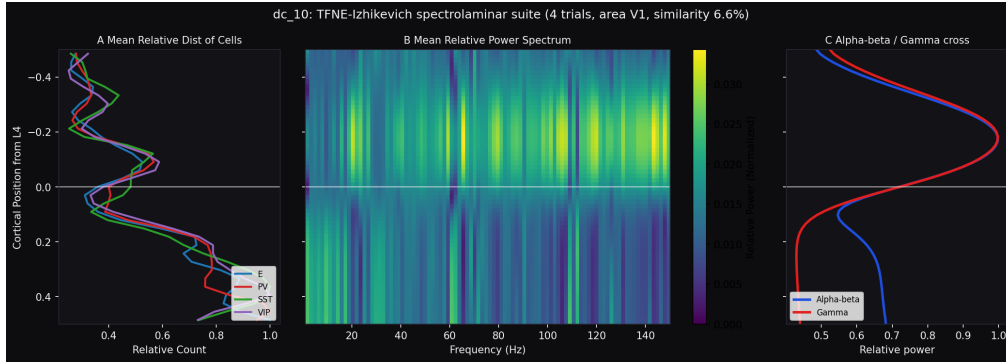


Figure 9: Spectrolaminar suite: $DC_{\text{deep}} = 10$ (strong deep drive). Note: despite increased drive to deep layers, the relative γ profile remains distributed across depths, not concentrated in superficial layers (see Table 2).

8 EEG and MEG proxies

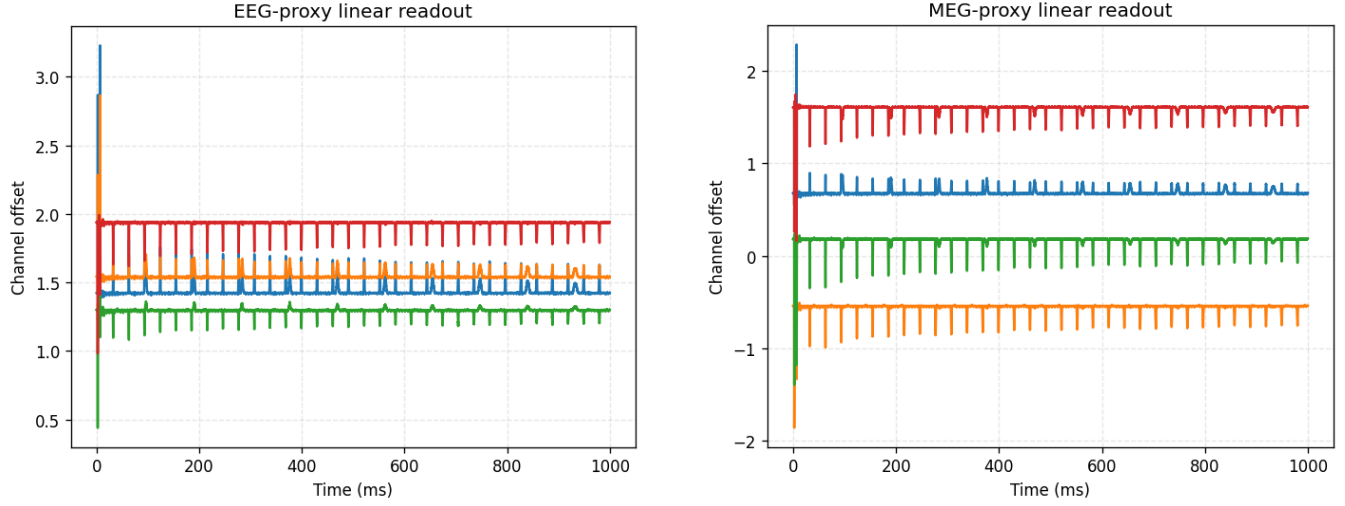


Figure 10: Left: `eeg_proxy` (linear leadfield projection to scalp electrodes). Right: `meg_proxy` (oriented-source magnetic projection). Both are proxy readouts in relative units.

9 Additional diagnostics

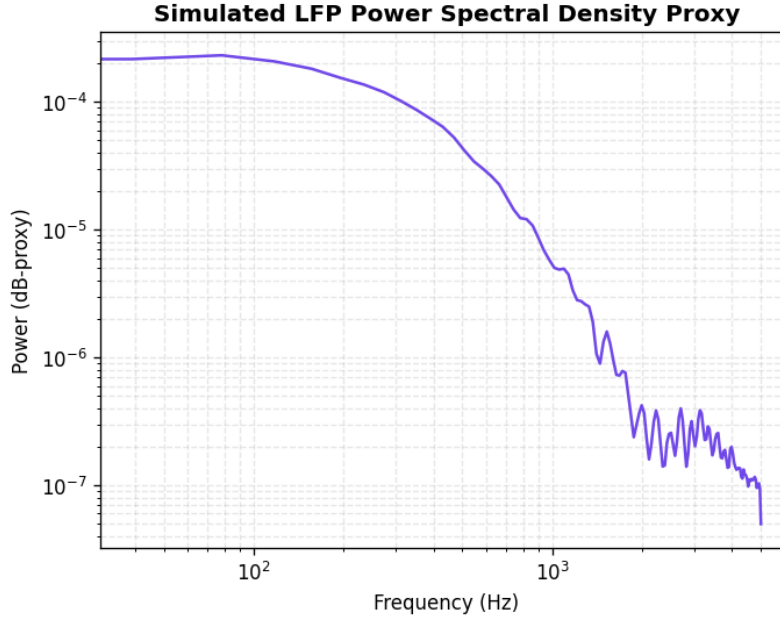


Figure 11: Power spectral density proxy of the laminar field.

10 Why the spectrolaminar suite is the preferred visualization

The spectrolaminar suite normalizes each band across depth, exposing *relative* laminar organization that raw LFP amplitude—dominated by the densest/most-driven layers—obscures, and it is robust to absolute-

amplitude scaling (appropriate here, as all outputs are uncalibrated proxies).

Scope note on the laminar crossing. The canonical spectrolaminar motif is α/β power peaking in deep layers and γ peaking superficially. In this scaffold that crossing is *readout- and scale-dependent*: it has been observed only via an explicit 128-contact *normalized* projection at the full $N = 10,000$ scale and does not transfer to reduced surrogates. On the suite’s default (16-contact) auto field used for the figures above, the measured band fractions (Table 2) do *not* reproduce it—deep γ fraction in fact rises with deep drive. We therefore report the monotonic rate/power increase as the robust effect and treat the laminar crossing as an unvalidated, scaffold-level observation, consistent with `claim_level=computational_scaffold`.

Scope statement

All field outputs are computational proxies (`claim_level=computational_scaffold`, `field_solver_status=linear_s`, `field_claim_level=proxy_readout`, `physical_amplitude_calibrated=False`). They are not physical EEG/MEG/LFP/CSD measurements and carry no calibrated amplitude.